

ISO/SAE 21434 TARA Threat Analysis

<https://gemini.google.com/app/123267e75d55d523>

User prompt: Act as a Senior Automotive and IoT Cybersecurity Engineer conforming to the ISO/SAE 21434 standard. Perform a preliminary Threat Analysis and Risk Assessment (TARA) — specifically Threat Identification and Damage Scenario formulation — for the system architecture modeled in Ansys Medini Analyze described below. ### System Architecture Overview The system is a multi-tier environmental/indoor monitoring platform featuring dual-path edge-to-cloud telemetry: Central Processing Unit: «Controller» Indoor Unit (IUC) Sensing Subsystem (connected to IUC): Particulate matter: PM2.5, PM10 Toxic/ambient gases: CO, CO2, NO2, SO2, O3 Geolocation: «Component» GPS (bidirectional interface with IUC) Actuation Subsystem: «Actuator» Status Display (controlled directly by IUC via Display In/Out lines) Uplink Path A (Local Edge / Cellular): IUC <-> «Component» Wi-Fi <-> «Controller» Far edge device <-> «Component» 4G <-> «Item/System» Cloud server Uplink Path B (LPWAN): IUC <-> «Component» LoRaWAN <-> «Component» LoRa GW <-> «Item/System» LoRa edge server <-> «Item/System» Cloud server Backend & Consumer Tier: «Item/System» Cloud server <-> «Item/System» User (dashboard / client application) Task Scope & Objectives 1. Apply the STRIDE threat modeling methodology systematically across: - Physical/analog sensor interfaces and buses to the IUC. - Local wireless channels (Wi-Fi, LoRaWAN, Cellular 4G, GPS RF signals). - Intermediate computing nodes (Far edge device, LoRa Gateway, LoRa edge server). - Cloud backend endpoints and end-user data delivery channels. 2. Formulate clear Threat Scenarios and map each to a concrete Damage Scenario considering vehicle/system-level safety, operation, financial impact, and user privacy (SFOP). ### Output Format Provide the results in a structured Markdown table compatible with Ansys Medini Analyze TARA tables, using the following columns: | Threat ID | Target Element / Interface | STRIDE Category | Compromised Asset & Property (Confidentiality, Integrity, Availability, Authenticity) | Threat Scenario Description | Damage Scenario (Vehicle / Passenger / Operational Impact) | |---|---|---|---|---|---| Ensure deep technical coverage of: - Sensor data spoofing/tampering (especially safety-critical gases like CO and toxic pollutants) causing suppressed alerts or false status indications on the Status Display. - Wireless jamming, replay, and eavesdropping attacks on LoRaWAN, 4G, Wi-Fi, and GPS. - Compromise or lateral movement via intermediate processing nodes (Far edge device, LoRa Edge server). Note: Please remember that there must not be any threads with more than one STRIDE category! One category per thread.

Response: Ansys Medini Analyze enforces strict 1:1 traceability between a **Threat Scenario** and a single **STRIDE Category**. The preliminary Threat Analysis and Risk Assessment (TARA) below conforms to **ISO/SAE 21434 Clause 15**, addressing safety, operational, financial, and privacy (SFOP) impacts across all architectural tiers.

Preliminary Threat Analysis & Risk Assessment (TARA) Table

Threat ID	Target Element / Interface	STRIDE Category	Compromised Asset & Property	Threat Scenario Description	Damage Scenario (Vehicle / Passenger / Operational Impact)
THR-01	Analog/Bus Line: Toxic Gas Sensors (CO) → IUC	Tampering	CO Measurement Data (Integrity)	An attacker physically taps or injects false analog voltage/PWM signals into the CO sensor line to suppress real CO level spikes.	[Safety / Operational] Driver and passengers are exposed to lethal Carbon Monoxide build-up without automated cabin ventilation or local visual warnings,

Threat ID	Target Element / Interface	STRIDE Category	Compromised Asset & Property	Threat Scenario Description	Damage Scenario (Vehicle / Passenger / Operational Impact)
					leading to severe illness or death.
THR-02	Display In/Out Line: IUC → Status Display	Spoofing	Display Command/Frame (Authenticity)	An attacker performs frame injection over the display bus interface, masquerading as the IUC to force a "SAFE / Air Quality OK" status.	[Safety / Operational] Critical visual alerts for hazardous gas levels (CO, SO ₂) are masked. Occupants remain in a toxic environment, leading to health emergencies.
THR-03	Interface: GPS → IUC	Spoofing	GPS NMEA Sentences / Location (Authenticity)	An adversary generates synthesized GNSS RF signals (GPS Spoofing) forcing the IUC to parse fake location coordinates.	[Operational / Financial] Fleet operators receive falsified spatial/environmental tracking data, triggering unnecessary emergency dispatches or failing to locate a compromised vehicle.
THR-04	Air Interface: GPS RF Signal → GPS Component	Denial of Service	GPS Signal Availability (Availability)	An attacker deploys a local RF jammer operating on the L1 frequency band (1575.42 MHz), blocking GPS signal reception.	[Operational] Telemetry logs lose spatial context; automated geo-fenced safety triggers (e.g., HVAC recirculate in tunnels) fail to execute.
THR-05	Air Interface: IUC ↔ Wi-Fi	Information Disclosure	Telemetry Stream in Transit (Confidentiality)	An attacker passively sniffs unencrypted or weakly encrypted Wi-Fi frames between the IUC and the Far Edge Device.	[Privacy] Driver travel patterns, cabin occupancy indicators, and micro-environmental habits are intercepted and exposed to unauthorized third parties.
THR-06	Interface: Wi-Fi Channel	Repudiation	Audit Logs / Transmission	A rogue local edge node clears or alters	[Financial / Legal] Incident forensically cannot attribute

Threat ID	Target Element / Interface	STRIDE Category	Compromised Asset & Property	Threat Scenario Description	Damage Scenario (Vehicle / Passenger / Operational Impact)
	(IUC ↔ Far Edge Device)		History (Repudiation)	transmission timestamps and packet sequence counters before forwarding data.	missing emergency alerts to network drops versus internal sensor failure during liability investigations.
THR-07	Local Wireless: LoRaWAN RF (IUC ↔ LoRa GW)	Repudiation	Downlink Ack / Frame Counter (Repudiation)	An attacker intercepts LoRaWAN uplinks and drops confirmation frames without dropping data frames, causing sync state mismatch.	[Operational] Disruption of dual-path uplink synchronization leading to redundant retransmissions, battery drain, and uncoordinated fallback logic.
THR-08	Node: Far Edge Device	Elevation of Privilege	Operating System / Root Privileges (Elevation of Privilege)	An attacker exploits an unpatched OS vulnerability on the Far Edge Device to execute arbitrary code with root privileges.	[Safety / Operational] Attacker gains control over edge processing, tampers with gas aggregation algorithms, and uses the edge node to pivot laterally into the IUC.
THR-09	Node: LoRa Edge Server	Tampering	LoRa MAC Payload Parsing (Integrity)	An attacker with access to the LoRa Edge Server tampers with incoming payloads before forwarding them over MQTT/HTTPS to the main cloud server.	[Operational] Inconsistent environmental readings between Uplink Path A and Path B, causing cloud anomaly engine failures and erroneous fleet diagnostics.
THR-010	Interface: LoRa GW ↔ LoRa Edge Server	Denial of Service	Backhaul Network Bandwidth (Availability)	An attacker floods the gateway-to-server backhaul link with invalid UDP/IP packet bursts, causing high packet loss.	[Operational] Complete failover dependency on Path A (Cellular 4G); if Path A fails, total blackout of edge-to-cloud environmental monitoring occurs.

Threat ID	Target Element / Interface	STRIDE Category	Compromised Asset & Property	Threat Scenario Description	Damage Scenario (Vehicle / Passenger / Operational Impact)
THR-011	Node: Cloud Server Endpoint	Elevation of Privilege	Server API / Database Access (Elevation of Privilege)	An attacker exploits an API authentication bypass on the Cloud Server to elevate privileges to System Administrator.	[Financial / Privacy] System-wide data breach compromising the personal data and historical location profiles of the entire vehicle fleet, exposing the OEM to GDPR fines.
THR-012	Interface: Cloud Server → User App	Spoofing	Push Notifications / Dashboard Data (Authenticity)	An attacker executes a Man-in-the-Middle (MitM) attack between the Cloud Server and User App using a rogue TLS certificate.	[Operational / Safety] Users receive false safety alerts or suppressed warnings on their mobile app dashboard, prompting unnecessary panics or unsafe driving decisions.

User prompt: i need you to tell me the total number of spoofing, tampering ... etc for all the stride letters. just count them for me and tell me how many for each actegory

Response: Here is the count of identified threat scenarios across each STRIDE category from the preliminary TARA analysis:

STRIDE Category Breakdown

STRIDE Category	Category Description	Threat IDs	Category Count
Spoofing	Impersonating an entity, sensor, or interface	THR-02, THR-03, THR-012	3
Tampering	Unauthorized modification of data or signals	THR-01, THR-09	2
Repudiation	Denying an action without auditable proof	THR-06, THR-07	2
Information Disclosure	Exposing sensitive data to unauthorized parties	THR-05	1

STRIDE Category	Category Description	Threat IDs	Category Count
Denial of Service	Preventing legitimate access or degrading availability	THR-04, THR-010	2
Elevation of Privilege	Gaining unauthorized higher-level permissions	THR-08, THR-011	2
Total Threat Scenarios			12